Temperature Stress' Effect on the Growth of Turions of *Spirodela Polyrhiza* Plants

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Abstract

The following experiment was conducted in order to study the effects of temperature stress on aquatic plant life, and how it can affect Earth's plant development. This study aimed to answer the question, "How does temperature stress affect the growth and development of *spirodela polyrhiza*?". It was predicted that higher levels of temperature stress would cause greater duckweed to grow at a slower rate. Photos were taken of the plant growth of three separate groups of greater duckweed that were each grown at different temperatures (20°C, 30°C, and 40°C) at three different time intervals: 7 days, 14 days, and 21 days. This data was analyzed by extracting the DNA from the plants on day 21 and running gel electrophoresis to get further information about the trends of each sample's growth. It was found that higher temperatures in fact do cause a slower growth rate in spirodela polyrhiza, and this information was added to the research done by several scientists over time. By completing this experiment and analyzing the results closely, this experiment was able to help support the various studies that other scientists have conducted on this same topic. This will ultimately help others to

understand the significance of temperature on plant growth and can help bring awareness to climate change and its possible effects on Earth.

Introduction:

Global climate change is a controversial environmental issue that the planet faces today. Climate change is caused by changes in several environmental factors such as temperature, air quality, food availability, and water quality and availability. It is these specific changes that cause drastic changes in epigenetics, phenotypes, and physiology of plants (Skinner, 2023). In recent years, climate change has had an effect on aquatic plant life, and coral reefs by altering plants' physiology and phenotypes (Skinner, 2023). Some examples of how aquatic plants' physiology and phenotypes are negatively affected include a reduction in biodiversity, the dominance of invasive species, and hindered growth. All of these negative factors negatively affect aquatic plant life as a whole. One method that plants use to respond to negative environmental impacts is epigenetic regulation. Epigenetic regulation is the alteration of gene expression in order to cause a cellular and organism response. It is the primary process involved with how the environment impacts living organisms (Skinner, 2023). On a more specific level, DNA methylation is an epigenetic mechanism that involves transferring a methylol group onto a C5 position of the cytosine in order to form 5-methylytosine (Moore *et al.* 2013). DNA methylation regulates the expression of genes by recruiting proteins that are involved in repressing certain genes or stopping the binding of transcription factors to DNA (Moore *et al.* 2013).

Temperature is a key ecological driver of plant functions and gene expression because it modifies membrane properties, the activity level of enzymes, the rate of several chemically affected reactions, the viscosity of vacuole solution and phloem, cytoplasm, and xylem solution found within plants (Źróbek-Sokolnik, 2011). These changes cause epigenetic modifications by changing chemical reactions and diffusions within plants. This is an environmental factor that

has one of the most influential effects of the physiology and gene expression of plants. This influences gene regulation without changing the DNA because when these stressors occur, the phenotype of the plant is affected, rather than the DNA and genes of the plant itself. An example of DNA methylation is the effect of high-temperatures on two protein-coding genes, At3g50770 and At5g43260 (Naydenov *et al.* 2015). Methylation mediates stress tolerance and phenotypic plasticity by acting as an extremely dynamic and not permanent mechanism that changes the expression of genes in response to different environmental factors (Naydenov *et al.* 2015). It mediates transgenerational inheritance by not directly affecting the DNA sequence of plants during its processes (Naydenov *et al.* 2015).

Spirodela polyrhiza (also known as greater duckweed) is a free-floating aquatic plant that has a fast growth rate and is found in many slow-moving or dormant bodies of freshwater (Jacobs, 1947). A few commonly known methylation patterns in duckweed include very low levels of methylation in the context of CHH, and a heavy reliance on maintenance methylation (Jacobs, 1947). These patterns align with its rapid growth rate. Greater duckweed is a plant that is very suitable for epigenetic studies because of its quick growth rate. This growth rate allows researchers to study different epigenetic affects across many generations of duckweed without genetic variation getting in the way of their work (Jacobs, 1947). A few prior findings on how temperature affects growth in duckweeds is that duckweed commonly grows at its fastest rate in water temperatures around 20° C and 30° C. In addition to this, in the past it has been found that high levels of heat stress on duckweed plants can affect their mortality rates (Jacobs, 1947).

A few gaps in current research on temperature-induced DNA methylation in aquatic plants is the overall long-term effect of climate change on aquatic plant life, and the mechanisms of inheritance. Many studies on this topic primarily focus on the short-term effects of

temperature on duckweed plants rather than the long term effect on them (Song *et al.* 2022). Methods that were used in the past to examine methylation responses and their strengths/weaknesses are high-resolution bisulfite sequencing, and Landmark Genomic Scanning (RLGS) (Song *et al.* 2022). Linking temperature stress to methylation patterns in spirodela is critical for understanding how aquatic plants adapt to climate change because by learning the effects that temperature have on this specific plant over time can help to bring light to the effects of climate change on several aquatic plants in today's oceans.

This study will be more closely focused on analyzing the long-term effects of temperature stress on these plants rather than temperature's short-term effects. The study that is conducted will fix some of these gaps in research by closely analyzing the change in these duckweed plants on the microscopic level by analyzing their DNA. This experiment will include groups at three different temperatures with each group containing six replicates. One turion pod of *Spirodela Polyrhiza* is added to each well containing Hogland's plant growth solution. The three groups will be kept at the temperatures of 20°C, 30°C, and 40°C over a course of 21 days of observation. The hypothesis concluded within this experiment is that the higher temperatures in this experiment will cause a slower growth rate of spirodela polyrhiza, and lower temperatures will cause a more rapid growth rate of spirodela polyrhiza. This hypothesis was decided upon because turions are a mechanism for duckweed plants to survive harsh winter conditions and still produce offspring. Turions sink to low water levels when temperatures get too cold or too hot for duckweed plants. Below the surface of the water, water temperatures are lower than they are on the surface. This is why it is believed that turions will grow at more rapid rates in colder water conditions. By conducting this experiment, research is being done that can potentially help to

discover more of the ways that climate change is affecting aquatic plants, and can potentially help create solutions for this global climate issue.

Methodology:

Experimental Design

For the preparation of this experiment, first, 10mL of sterile Hoagland's solution was dispensed into each well using a serological pipette. This solution was added in order to aid the growth of the plants that will be monitored. In total, 18 wells were filled. This experiment includes groups at three different temperatures with each group containing six replicates. After this, one turion pod of *Spirodela Polyrhiza* was added to each well containing the Hogland's solution. Each pod was added using a sterile inoculation loop. In between adding each pod, the loop was sterilized using the flame of a Bunsen burner to ensure that this experiment was as sterile as possible. After adding all the turion pods to the wells, each well was labeled with the number 1-18 with a sharpie, and each microplate was labeled with the names of each group member, the section number of the lab, and the temperatures at which each group will be stored (20°C, 30°C, and 40°C) using a piece of tape.

Data Collection:

To collect data, first, photos were taken of each group of replicates and uploaded onto the computer being used for the experiment as jpeg images. Following this, the online application ImageJ was used to upload the photos and analyze them. Each point was cropped to focus on one well at a time. Each duckweed frond was counted using the "Multipoint" tool in ImageJ. The number of fronds in each well was recorded into an excel spreadsheet under the column labeled, "Frond Count". After this, the "Straight" tool was used to draw a line that represented the diameter of each well. Using the "Set Scale" feature in ImageJ, the size of each well was set by adding the known distance as 34.8, and the units as

millimeters. This was done for every well that was analyzed. Finally, the “Freehand” tool was used to trace the outline of the duckweed fronds in each well. Using the “Measure” feature, ImageJ automatically measured the area of the fronds that were being observed. This area was recorded in the excel spreadsheet under the column labeled, “Duckweed Area”. After adding this value, the column labeled, “Percent Coverage” was automatically updated because it was pre-set with the formula, Duckweed Area / Well Area *100.

Genetic Sequencing:

To begin, each group put on personal protective equipment. Following this, each group member obtained a plastic tube rack and placed the necessary materials into it. The materials that were collected were M-Binding Buffer, M-Wash Buffer, M-Desulphonation Buffer, and M-Elution Buffer. Two collection tubes were also acquired and labeled with E, for Elute, and D, for discard. A Microcentrifuge Tube was collected and labeled B, for Bind. A Zymo-Spin column was collected and finally, 2 empty PCR Tubes were collected and labeled with F, for final, and X, for extra. To begin, 400 microliters of the M-Binding Buffer were added to the Zymo-Spin Column that was acquired. 150 microliters of the DNA sample that each group member previously collected was also added. This tube was inverted to mix its contents, and centrifuged for 30 seconds. The flow-through was discarded into the D tube. These steps were repeated with slightly different amounts of each solution, and was repeated twice with the M-Wash Buffer. The lats step consisted of adding 10 microliters of M-Elution Buffer directly to the center of the column membrane. It was centrifuged for 30 seconds and then one microliter of the resulting final solution was added to the F and X tubes and given to the lab professor. These steps were followed in order to clean the DNA that was acquired during our experiment to ensure that a clean DNA sample was available to analyze.

Data Analysis:

Within this experiment, frond count and percent coverage of *spirodela* polyrhiza was measured for growth. The statistical test used was an Analysis of Variance (ANOVA) test which explained whether or not the results that were gained during the experiment were significant or not. It was run in a Python library. The post-hoc test that was used was a Tukey Test and this test was completed to determine how the treatments were different from each other. If the p-value in the ANOVA test was less than 0.05, those findings would be considered significant. The software that was used to generate the figures and tables was Microsoft excel. The post-hoc test that was used to compare methylation between treatments was a Tukey-Test.

Results

Spirodela polyrhiza frond number increased significantly between the heat-stressed and non heat-stressed treatments across the experimental period (Fig. 1). Plants in the 20° and 30° treatments produced more fronds than those in the 40° treatment by day 9. These results suggest that higher temperatures negatively affect duckweed growth. This is because the higher the temperature that the fronds were placed in, the less growth that the duckweed had.

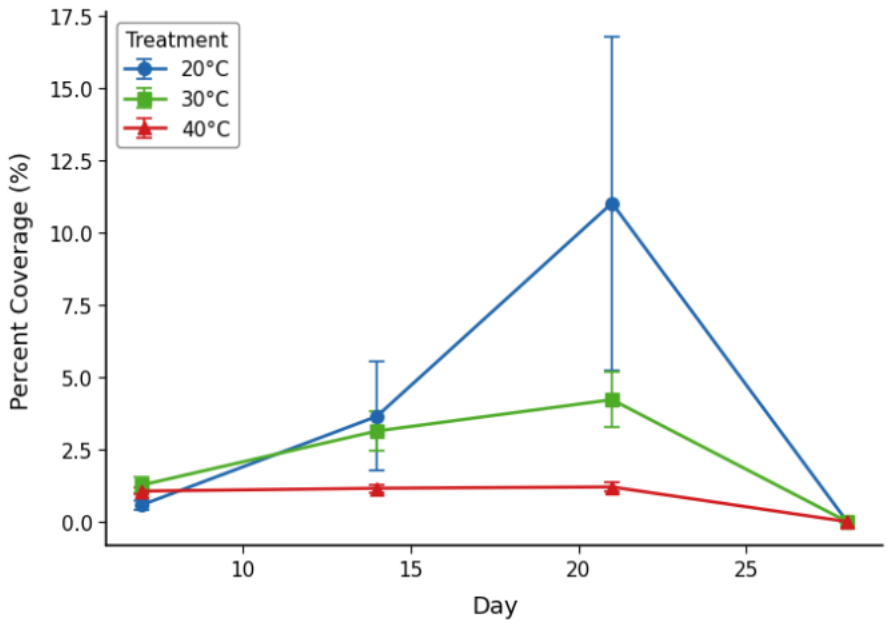


Figure 1. Percent coverage between Treatments over time. Each color represents a different temperature treatment, and the percent coverage (shown on the left) is tracked for each day (shown on the bottom of the graph). Red represents 40°C, green represents 30°C, and blue represents 20°C.

Table 1. Turkey Test: GROWTH – Percent Coverage between Treatments. Turkey tests show which experimental groups differ from each other. Groups 1 and 2 are the groups being compared, the meandiff is the difference between the means of both groups, the p-adj is the p-value but adjusted for several different comparisons, and the "True" under the reject column means that the difference is statistically significant ($p<0.05$), and "False" means that it is not.

Group 1	Group 2	meandiff	p-adj	lower	upper	reject
20°C	30°C	-1.6549	0.4812	-5.0732	1.7634	False
20°C	40°C	-2.9577	0.1031	-6.3759	0.4606	False
30°C	40°C	-1.3028	0.634	-4.721	2.115	False

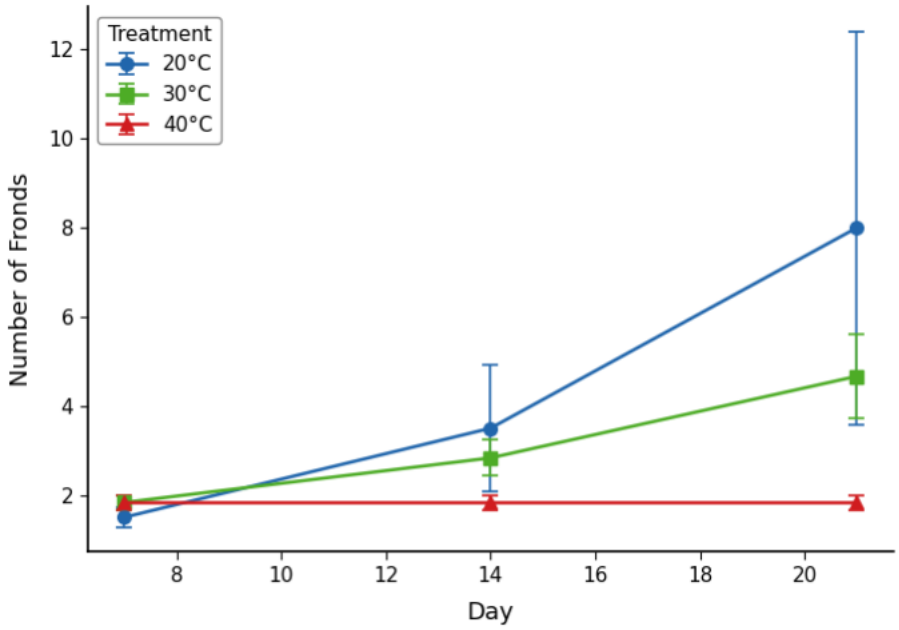


Figure 2. Number of Fronds between Treatments over time. Each color represents a different temperature treatment, and the number of fronds (shown on the left) is tracked for each day (shown on the bottom of the graph). Red represents 40°C, green represents 30°C, and blue represents 20°C.

Table 2. Turkey Test: GROWTH – Number of Fronds between Treatments. Turkey tests show which experimental groups differ from each other. Groups 1 and 2 are the groups being compared, the meandiff is the difference between the means of both groups, the p-adj is the p-value but adjusted for several different comparisons, and the "True" under the reject column means that the difference is statistically significant ($p<0.05$), and "False" means that it is not.

Group 1	Group 2	meandiff	p-adj	lower	upper	reject
20°C	30°C	nan	nan	nan	nan	False
20°C	40°C	nan	nan	nan	nan	False
30°C	40°C	nan	nan	nan	nan	False

Heat-stressed plants showed a methylated band at the HSP70 locus (Fig. 3), while control plants did not amplify this product. This pattern suggests that temperature stress negatively affects DNA methylation in Spirodela polyrhiza by hindering the process of DNA methylation when temperatures rise. This is likely due heat-stress' affect on duckweed growth. Because duckweed grows less when exposed to heat-stress, DNA methylation is affected as a result.

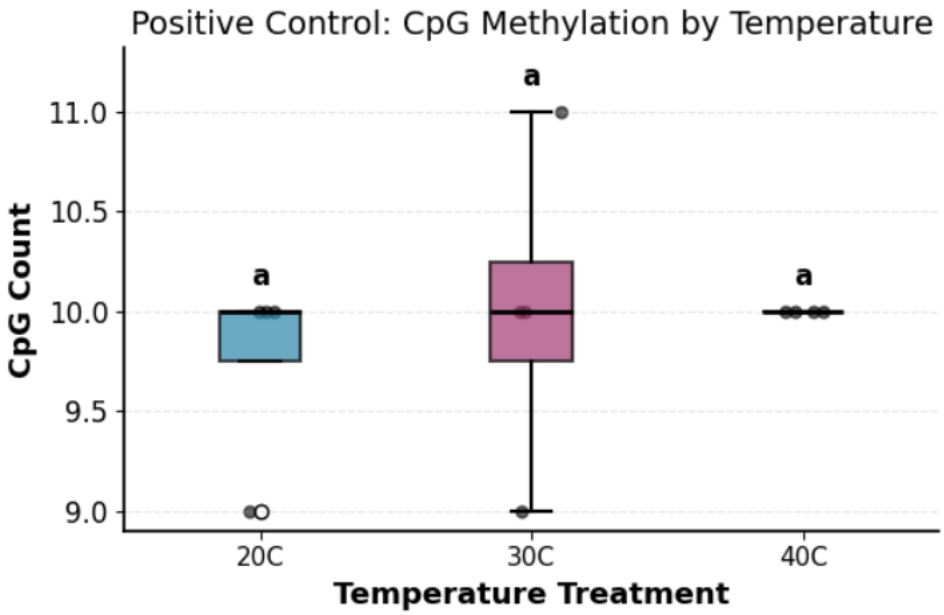
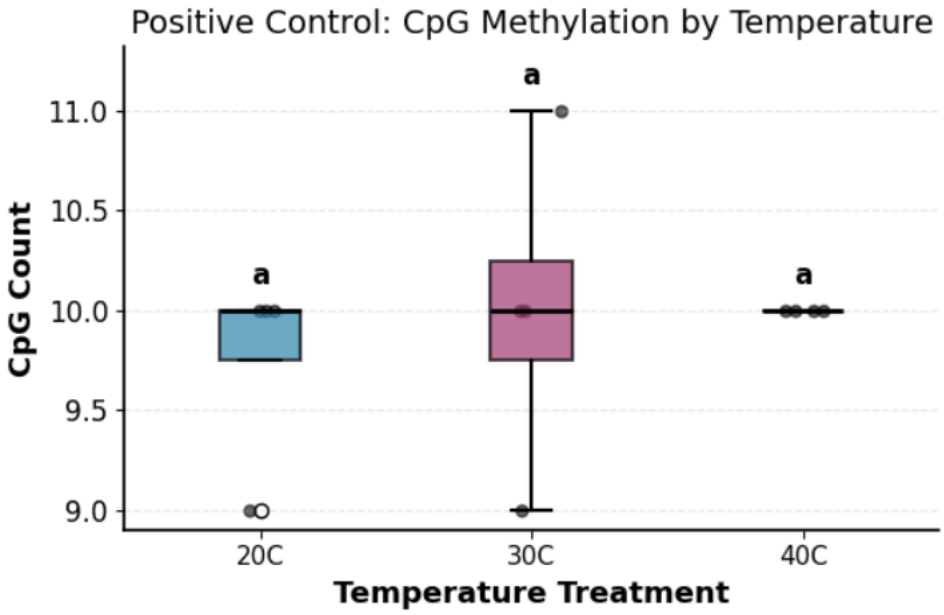


Figure 3. Endpoint Methylation Comparison. The left side of the graph shows CpG count, while the bottom of the graph shows which Temperature Treatments are being compared. Each number on the CpG section of the graph is a representation of how many methylated CpG sites were detected in the amplified region of each gene.

Table 3. Turkey Test METHYLATION – CpG Sites between Treatments. The figure shows CpG Count (represented on the left of the graph) and temperature treatments of the experiment (represented on the bottom of the graph). The colored bars in the center of the graph represent the CpG count of the specific temperature treatments of our experiment.



Discussion

The null hypothesis of this experiment expressed that heat stress on the spirodela polyrhiza samples will have no effect on their growth. In contrast, the alternative hypothesis was that the higher temperatures in this experiment will cause a slower growth rate of spirodela polyrhiza, and lower temperatures will cause a more rapid growth rate of spirodela polyrhiza. Overall, the experiment results showed that the higher the heat stress was on a sample of spirodela polyrhiza, the slower its growth rate was. This data led to the rejection of the null hypothesis.

Heat stress most likely reduces growth in duckweed because duckweed does not properly grow in conditions with heat stress. In addition to this, turions are adapted in a way that helps duckweed grow despite very cold climates. When exposed to heat, they show less growth because they are adapted to cold temperatures rather than hot temperatures. Heat stress most likely changes DNA methylation because methylation mediates stress tolerance and phenotypic plasticity by acting as an extremely dynamic and not permanent mechanism that changes the expression of genes in response to different environmental factors (Naydenov et al. 2015).

An unexpected result that was encountered was found in the day 21 results of the 20° samples. There were a few wells in this group that had no growth at all. This was surprising because the pattern that had been studied up to this point was that the 20° wells contained the most duckweed growth. One limitation of this study was the growth period. If the experiment had a longer growth period to analyze more prolonged results, more data could have been analyzed which would have been beneficial in research for this experiment. Another limitation that was faced was data interpretation. Because this was the entire group's first time working with duckweed, the interpretations of this experiment could potentially be looked at incorrectly due to inexperience.

A question that the results of this experiment raised that this study could not answer was what the results would be if the duckweed samples grew for longer periods of time. If a change were to be made to this experiment in the future, the duckweed samples would be allowed to grow for a longer period of time in order to see more detailed trends in the growth of the samples over a prolonged period. The technique that would be used in this scenario would be the same techniques that were used within this experiment: Growing duckweed fronds, analyzing their growth over a prolonged period of time, and finally running tests such as ANOVA and Tukey Tests in order to measure significance of the results. By doing this, more detailed results would be expected to be found, which would ultimately benefit the accuracy of this experiment overall.

If temperature stress can affect duckweed growth, it can affect the growth of other plants as well. This is imperative information to be aware of because climate change affects the world every day, and it would be beneficial to know how these increased temperatures can affect other plants that are commonly used on Earth. Understanding how aquatic plants respond to thermal stress at the molecular level is important because it can show how thermal stress is able to affect other plants throughout the world, which in turn leads to how humans in general are affected.

References:

Jacobs, D. L. (1947). An Ecological Life-History of Spirodela Polyrhiza (Greater Duckweed) with Emphasis on the Turion Phase. *Ecological Monographs*, *17*(4), 437–469. https://doi.org/10.2307/1948596

Moore, L. D., Le, T., & Fan, G. (2013). DNA methylation and its basic function. *Neuropsychopharmacology*, *38*(1), 23-38.

Naydenov, M., Baev, V., Apostolova, E., Gospodinova, N., Sablok, G., Gozmanova, M., & Yahubyan, G. (2015). High-temperature effect on genes engaged in DNA methylation and affected by DNA methylation in Arabidopsis. *Plant Physiology and Biochemistry*, *87*, 102-108.

Skinner, M. K. (2023). Environmental epigenetics and climate change. *Environmental epigenetics*, *9*(1), dvac028.

Song, H., Guo, X., Liu, L., Xu, Z., Wang, N., Liu, X., ... & Guo, W. (2022). Role of DNA methylation in ecophysiological responses to salinity in natural populations of Phragmites australis from coastal and inland habitats. *Frontiers in Marine Science*, *9*, 982234.

Źróbek-Sokolnik, A. (2011). Temperature stress and responses of plants. In *Environmental adaptations and stress tolerance of plants in the era of climate change* (pp. 113-134). New York, NY: Springer New York.